Domain : AGRITECH

Project Title:Cold Storage Temperature Monitoring System-Uttarpradesh

⚠️ Limitations of Current System

1. Basic Alert Logic

- Alerts are triggered only when temperature exceeds a **fixed threshold (10°C)**.
- No dynamic thresholds for different crops (e.g., potatoes vs fruits).
- No multi-level alerts (warning, critical, emergency).

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2. No Real IoT Integration

- Temperature data is **manually entered**.
- No connection with actual sensors like Arduino, ESP32, or IoT devices.
- Not suitable for real-time industrial deployment yet.

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3. Limited Database Capability

- Uses **SQLite**, which is:
  - Not ideal for large-scale systems
  - Lacks high concurrency support
- No data backup or failover mechanism.

🚀 Future Enhancements

(Cold Storage Temperature Monitoring System)

1. IoT-Based Automatic Temperature Monitoring

- Integrate sensors like:
  - DHT11 / DHT22
  - ESP32 / Arduino
- Enable **real-time automatic data collection** without manual entry.
- Improves accuracy and reliability.

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2. Smart Alert System 🔔

- Implement **multi-level alerts**:
  - Warning (slight deviation)
  - Critical (high risk)
  - Emergency (immediate action)
- Add:
  - SMS alerts
  - Email notifications
  - Mobile push notifications

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3. Crop-Specific Temperature Control 🌾

- Different crops require different temperature ranges:
  - Potatoes: 2–4°C
  - Fruits: 0–2°C
- Allow users to:
  - Select crop type

    - Automatically set safe temperature thresholds

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4. Advanced Analytics & AI 📊

- Add intelligent features:
    - Predict temperature trends
    - Detect abnormal patterns
    - Spoilage risk prediction
- Use basic Machine Learning models for forecasting.

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5. Cloud Deployment ☁️

- Deploy system on:
    - AWS / Azure / Render
- Enable:
    - Remote monitoring from anywhere
    - Data backup and recovery
    - High availability

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6. Multi-Storage Unit Management 🏢

- Fully implement:
    - Multiple cold storage units
    - Separate dashboards for each unit
- Add centralized monitoring system.

. Power Backup & Stability ⚡

- Cold storages often face power cuts.
- Must include:
  - UPS (Uninterruptible Power Supply)
  - Backup generators
- Ensure system continues:
  - Monitoring
  - Logging data during outages

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3. Network Connectivity 🌐

- Rural areas may have poor internet.
- Consider:
  - GSM/4G modules
  - Offline data storage with later sync
- Ensure continuous communication for alerts.

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4. Scalability 📈

- System should support:
  - Multiple storage units
  - Large number of sensors
- Use scalable technologies:
  - Cloud databases (instead of SQLite)
  - Microservices architecture (future scope)

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5. Data Security & Privacy 🔐

- Protect sensitive data:
  - Storage conditions
  - Business records
- Implement:
  - User authentication
  - Encryption (HTTPS)
  - Secure APIs

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6. Real-Time Alert Reliability 🔔

- Alerts must be:
  - Instant
  - Reliable
- Use multiple channels:
  - SMS
  - Email
  - App notifications
- Add redundancy (send alerts via multiple methods).

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7. Maintenance & Support 🛠️

- Regular system checks required:
  - Sensor calibration
  - Software updates
- Provide:

    - Easy maintenance interface
    - Logs for troubleshooting

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8. User Training 👩‍🏭

- Workers may not be tech-savvy.
- Provide:
    - Simple UI
    - Training sessions
    - Manuals in regional languages

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9. Environmental Factors 🌡️

- Cold storage conditions:
    - Low temperatures
    - High humidity
- Use hardware designed for harsh environments